

D. x^2

4. L&E, 2ADV E1 2013 HSC 9 MC

What is the solution of $5^x = 4$?

- A. $x = \frac{\log_2 4}{5}$
- B. $x = \frac{4}{\log_2 5}$
- C. $x = \frac{\log_2 4}{\log_2 5}$
- D. $x = \log_2 \left(\frac{4}{5} \right)$

5. L&E, 2ADV E1 2023 HSC 8 MC

What is the solution of the equation $\log_a x^3 = b$, where a and b are positive constants?

- A. $x = b^{\frac{a}{3}}$
- B. $x = a^{\frac{b}{3}}$
- C. $x = \frac{b^a}{3}$
- D. $x = \frac{a^b}{3}$

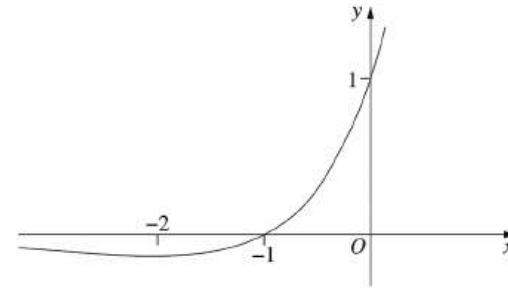
6. L&E, 2ADV E1 2014 HSC 3 MC

What is the solution to the equation $\log_2(x - 1) = 8$?

- A. 4
- B. 17
- C. 65
- D. 257

7. L&E, 2ADV E1 2015 HSC 8 MC

The diagram shows the graph of $y = e^x(1 + x)$.



How many solutions are there to the equation $e^x(1 + x) = 1 - x^2$?

- A. 0
- B. 1
- C. 2
- D. 3

8. L&E, 2ADV E1 2017 HSC 5 MC

It is given that $\ln a = \ln b - \ln c$, where $a, b, c > 0$.

Which statement is true?

- A. $a = b - c$
- B. $a = \frac{b}{c}$
- C. $\ln a = \frac{b}{c}$
- D. $\ln a = \frac{\ln b}{\ln c}$

9. L&E, 2ADV E1 2019 HSC 3 MC

What is the value of p so that $\frac{a^2 a^{-3}}{\sqrt{a}} = a^p$?

- A. -3
- B. $-\frac{3}{2}$
- C. $-\frac{1}{2}$
- D. 12

10. L&E, 2ADV E1 2019 HSC 5 MC

Which of the following is equal to $\frac{\log_2 9}{\log_2 3}$?

- A. 2
- B. 3
- C. $\log_2 3$
- D. $\log_2 6$

11. L&E, 2ADV E1 SM-Bank 11 MC

Solve the equation $e^{4x} - 5e^{2x} + 4 = 0$ for x

- A. $x = 1, 4$
- B. $x = -4, -1$
- C. $x = 0, \log_e 2$
- D. $x = -\log_e 2, 0, \log_e 2$

12. L&E, 2ADV E1 SM-Bank 6 MC

The expression

$$\log_c(a) + \log_a(b) + \log_b(c)$$

is equal to

- A. $\frac{1}{\log_c(a)} + \frac{1}{\log_a(b)} + \frac{1}{\log_b(c)}$
- B. $\frac{1}{\log_a(c)} + \frac{1}{\log_b(a)} + \frac{1}{\log_c(b)}$
- C. $-\frac{1}{\log_a(b)} - \frac{1}{\log_b(c)} - \frac{1}{\log_c(a)}$
- D. $\frac{1}{\log_a(a)} + \frac{1}{\log_b(b)} + \frac{1}{\log_c(c)}$

13. L&E, 2ADV E1 2006 HSC 1a

Evaluate $e^{-0.5}$ correct to three decimal places. (2 marks)

14. Functions, 2ADV F1 SM-Bank 53

i. If $\frac{1}{\sqrt[3]{7+\pi}} = (7+\pi)^x$, find x . (1 mark)

ii. Calculate the value of $\frac{1}{\sqrt[3]{7+\pi}}$ to 3 significant figures. (1 mark)

15. L&E, 2ADV E1 SM-Bank 8

Solve the equation $3^{-4x} = 9^{6-x}$ for x . (2 marks)

16. L&E, 2ADV E1 SM-Bank 12

Solve the equation $\log_e(3x+5) + \log_e(2) = 2$, for x . (2 marks)

17. L&E, 2ADV E1 2011 HSC 1c

Solve $2^{2x+1} = 32$. (2 marks)

18. L&E, 2ADV E1 2005 HSC 5a

Use the change of base formula to evaluate $\log_3 7$, correct to two decimal places. (1 mark)

19. L&E, 2ADV E1 SM-Bank 7

Solve the equation $2\log_3(5) - \log_3(2) + \log_3(x) = 2$ for x . (2 marks)

20. L&E, 2ADV E1 SM-Bank 9

Solve $\log_2(6-x) - \log_2(4-x) = 2$ for x , where $x < 4$. (2 marks)

21. L&E, 2ADV E1 SM-Bank 10

Solve the equation $2^{3x-3} = 8^{2-x}$ for x . (2 marks)

22. L&E, 2ADV E1 2009 HSC 1f

Solve the equation $\ln x = 2$. Give you answer correct to four decimal places. (2 marks)

23. L&E, 2ADV E1 2010 HSC 4d

Let $f(x) = 1 + e^x$.

Show that $f(x) \times f(-x) = f(x) + f(-x)$. (2 marks)

24. L&E, 2ADV E1 2023 MET1 2

Solve $e^{2x} - 12 = 4e^x$ for $x \in R$. (3 marks)

25. L&E, 2ADV E1 EQ-Bank 5

Solve the equation $8^{n+3} = \frac{1}{2}$ (2 marks)

26. L&E, 2ADV E1 SM-Bank 13

Find x given $100^{x-2} = 1000^x$. *(2 marks)*

27. L&E, 2ADV E1 2004 HSC 6a

Solve the following equation for x :

$$e^{2x} + 3e^x - 10 = 0. \text{ (2 marks)}$$

28. L&E, 2ADV E1 2007 HSC 6a

Solve the following equation for x :

$$2e^{2x} - e^x = 0. \text{ (2 marks)}$$

29. L&E, 2ADV E1 2008 HSC 7a

Solve $\log_e x - \frac{3}{\log_e x} = 2$ *(3 marks)*

30. L&E, 2ADV E1 2019 HSC 15a

Solve $e^{2\ln x} = x + 6$ *(2 marks)*

31. L&E, 2ADV E1 2020 MET1 4

Solve the equation $2\log_2(x + 5) - \log_2(x + 9) = 1$. *(3 marks)*

32. L&E, 2ADV E1 EQ-Bank 2

Show $f(x) = \frac{1}{2} - \frac{1}{2^x + 1}$ is an odd function. *(3 marks)*

33. L&E, 2ADV E1 EQ-Bank 3

Solve the equation $4^x - 2^{x+2} = 32$, showing all working. *(3 marks)*

34. L&E, 2ADV E1 SM-Bank 15

Solve the following equation for x :

$$2^{2x} = 3(2^{x+1}) - 8. \text{ (3 marks)}$$

35. L&E, 2ADV E1 2016 HSC 14e

Write $\log 2 + \log 4 + \log 8 + \dots + \log 512$ in the form $a \log b$ where a and b are integers greater than 1. *(2 marks)*

36. L&E, 2ADV E1 2019 MET1-N 4

Solve $\log_3(t) - \log_3(t^2 - 4) = -1$ for t . *(3 marks)*

Worked Solutions

1. L&E, 2ADV E1 SM-Bank 2 MC

$$\begin{aligned}y-3 &= \log_a(7x-b) \\ a^{y-3} &= 7x-b \\ a^{y-3} + b &= 7x \\ \therefore x &= \frac{1}{7}(a^{y-3} + b) \\ \Rightarrow C\end{aligned}$$

2. L&E, 2ADV E1 SM-Bank 4 MC

$$\begin{aligned}f(5x) &= 3\log_e(2(5x)) \\ \log_e(y) &= 3\log_e(10x) \\ &= \log_e(10x)^3 \\ y &= 1000x^3 \\ \Rightarrow D\end{aligned}$$

3. L&E, 2ADV E1 2012 HSC 7 MC

$$\begin{aligned}\log_e(a^2) &= \log_e(e^x)^2 \\ &= \log_e(e^{2x}) \\ &= 2x\log_e e \\ &= 2x \\ \Rightarrow C\end{aligned}$$

4. L&E, 2ADV E1 2013 HSC 9 MC

$$\begin{aligned}5^x &= 4 \\ \log_2 5^x &= \log_2 4 \\ x \log_2 5 &= \log_2 4 \\ \therefore x &= \frac{\log_2 4}{\log_2 5} \\ \Rightarrow C\end{aligned}$$

Worked Solutions

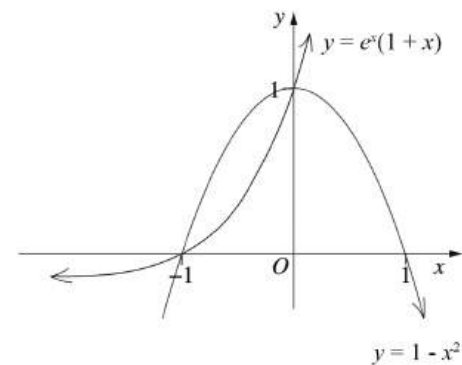
5. L&E, 2ADV E1 2023 HSC 8 MC

$$\begin{aligned}\log_a x^3 &= b \\ 3\log_a x &= b \\ \log_a x &= \frac{b}{3} \\ \therefore x &= a^{\frac{b}{3}} \\ \Rightarrow B\end{aligned}$$

6. L&E, 2ADV E1 2014 HSC 3 MC

$$\begin{aligned}\log_2(x-1) &= 8 \\ x-1 &= 2^8 \\ x &= 257 \\ \Rightarrow D\end{aligned}$$

7. L&E, 2ADV E1 2015 HSC 8 MC



The graphs intersect at 2 points

$$\begin{aligned}\therefore e^x(1+x) &= 1-x^2 \text{ has 2 solutions} \\ \Rightarrow C\end{aligned}$$

8. L&E, 2ADV E1 2017 HSC 5 MC

$$\begin{aligned}\ln a &= \ln b - \ln c \\ \ln a &= \ln\left(\frac{b}{c}\right) \\ \therefore a &= \frac{b}{c} \\ \Rightarrow B\end{aligned}$$

Mean mark 51%.
COMMENT: Use of log laws here proved difficult for many students.

9. L&E, 2ADV E1 2019 HSC 3 MC

$$\begin{aligned}\frac{a^2a^{-3}}{a^{\frac{1}{2}}} &= a^{-1} \times a^{-\frac{1}{2}} \\ &= a^{-\frac{3}{2}} \\ \Rightarrow B\end{aligned}$$

10. L&E, 2ADV E1 2019 HSC 5 MC

$$\begin{aligned}\frac{\log_2 9}{\log_2 3} &= \frac{\log_2 3^2}{\log_2 3} \\ &= \frac{2\log_2 3}{\log_2 3} \\ &= 2 \\ \Rightarrow A\end{aligned}$$

11. L&E, 2ADV E1 SM-Bank 11 MC

$$\begin{aligned}e^{4x}-5e^{2x}+4 &= 0 \\ \text{Let } X &= e^{2x} \\ X^2-5X+4 &= 0 \\ (X-4)(X-1) &= 0 \\ X &= 4 \text{ or } 1 \\ \therefore e^{2x} &= 4 & e^{2x} &= 1 \\ 2x &= \log_e 4 & x &= 0 \\ x &= \frac{2\log_e 2}{2} \\ &= \log_e 2 \\ \Rightarrow C\end{aligned}$$

12. L&E, 2ADV E1 SM-Bank 6 MC

$$\begin{aligned}\text{Solution 1} \\ \text{Using Change of Base:} \\ \log_c(a) + \log_a(b) + \log_b(c) \\ &= \frac{\log_a(a)}{\log_a(c)} + \frac{\log_b(b)}{\log_b(a)} + \frac{\log_c(c)}{\log_c(b)} \\ &= \frac{1}{\log_a(c)} + \frac{1}{\log_b(a)} + \frac{1}{\log_c(b)} \\ \Rightarrow B\end{aligned}$$

Solution 2

$$\begin{aligned}\text{Let } x &= \log_c(a) \\ c^x &= a \\ x\log_a c &= \log_a a \\ x &= \frac{1}{\log_a c}\end{aligned}$$

$$\begin{aligned}\text{Apply similarly for the other terms.} \\ \Rightarrow B\end{aligned}$$

13. L&E, 2ADV E1 2006 HSC 1a

$$\begin{aligned}e^{-0.5} &= 0.6065\dots \\ &= 0.607 \text{ (to 3 d.p.)}\end{aligned}$$

14. Functions, 2ADV F1 SM-Bank 53

$$\begin{aligned}\text{i. } \frac{1}{\sqrt[3]{7+\pi}} &= (7+\pi)^{-\frac{1}{3}} \\ \text{ii. } \frac{1}{\sqrt[3]{7+\pi}} &= 0.4619\dots \\ &= 0.462 \text{ (to 3 sig. fig.)}\end{aligned}$$

15. L&E, 2ADV E1 SM-Bank 8

$$\begin{aligned}3^{-4x} &= (3^2)^{6-x} \\ 3^{-4x} &= 3^{12-2x} \\ -4x &= 12-2x \\ 2x &= -12 \\ \therefore x &= -6\end{aligned}$$

16. L&E, 2ADV E1 SM-Bank 12

Simplify using log laws:

$$\begin{aligned}\log_e(6x + 10) &= 2 \\ 6x + 10 &= e^2 \\ \therefore x &= \frac{e^2 - 10}{6}\end{aligned}$$

17. L&E, 2ADV E1 2011 HSC 1c

$$\begin{aligned}2^{2x+1} &= 32 \\ 2^{2x+1} &= 2^5 \\ 2x + 1 &= 5 \\ \therefore x &= 2\end{aligned}$$

MARKER'S COMMENT: Many students correctly solved this by taking the logarithms of both sides.

18. L&E, 2ADV E1 2005 HSC 5a

$$\begin{aligned}\log_3 7 &= \frac{\log_{10} 7}{\log_{10} 3} \\ &= 1.771\dots \\ &= 1.77 \text{ (to 2 d.p.)}\end{aligned}$$

19. L&E, 2ADV E1 SM-Bank 7

$$\begin{aligned}\log_3(5)^2 - \log_3(2) + \log_3(x) &= 2 \\ \log_3(25x) - \log_3(2) &= 2 \\ \log_3\left(\frac{25x}{2}\right) &= 2 \\ \frac{25x}{2} &= 3^2 \\ \therefore x &= \frac{18}{25}\end{aligned}$$

20. L&E, 2ADV E1 SM-Bank 9

Simplify using log laws:

$$\begin{aligned}\log_2\left(\frac{6-x}{4-x}\right) &= 2 \\ 2^2 &= \frac{6-x}{4-x} \\ 16 - 4x &= 6 - x \\ 3x &= 10 \\ \therefore x &= \frac{10}{3}\end{aligned}$$

21. L&E, 2ADV E1 SM-Bank 10

$$\begin{aligned}2^{3x-3} &= 2^{3(2-x)} \\ 3x - 3 &= 6 - 3x \\ 6x &= 9 \\ \therefore x &= \frac{3}{2}\end{aligned}$$

22. L&E, 2ADV E1 2009 HSC 1f

$$\begin{aligned}\ln x &= 2 \\ \log_e x &= 2 \\ x &= e^2 \\ &= 7.38905\dots \\ &= 7.3891 \text{ (to 4 d.p.)}\end{aligned}$$

MARKER'S COMMENT: Students are reminded to write answers to more decimal places than required before rounding up.

23. L&E, 2ADV E1 2010 HSC 4d

$$\begin{aligned}f(x) \times f(-x) &= (1 + e^x)(1 + e^{-x}) \\ &= 1 + e^{-x} + e^x + e^x e^{-x} \\ &= e^x + e^{-x} + 2\end{aligned}$$

MARKER'S COMMENT: A common error in this question was not to realise that $e^x e^{-x} = e^0 = 1$.

$$\begin{aligned}f(x) + f(-x) &= 1 + e^x + 1 + e^{-x} \\ &= e^x + e^{-x} + 2 \\ &= f(x) \times f(-x) \dots \text{as required}\end{aligned}$$

24. L&E, 2ADV E1 2023 MET1 2

$$\begin{aligned}e^{2x} - 12 &= 4e^x \\e^{2x} - 4e^x - 12 &= 0\end{aligned}$$

Let $u = e^x$:

$$\begin{aligned}u^2 - 4u - 12 &= 0 \\(u - 6)(u + 2) &= 0\end{aligned}$$
$$\Rightarrow u = 6 \text{ or } -2$$
$$\therefore e^x = 6 \quad \text{or} \quad e^x = -2 \text{ (no solution)}$$
$$x = \log_e 6$$

25. L&E, 2ADV E1 EQ-Bank 5

$$\begin{aligned}8^{n+3} &= \frac{1}{2} \\2^{3(n+3)} &= 2^{-1} \\3n + 9 &= -1 \\3n &= -10 \\n &= -\frac{10}{3}\end{aligned}$$

26. L&E, 2ADV E1 SM-Bank 13

$$\begin{aligned}100^{x-2} &= 1000^x \\(10^2)^{x-2} &= (10^3)^x \\10^{2x-4} &= (10)^{3x} \\2x - 4 &= 3x \\\therefore x &= -4\end{aligned}$$

27. L&E, 2ADV E1 2004 HSC 6a

$$\begin{aligned}e^{2x} + 3e^x - 10 &= 0 \\\therefore (e^x)^2 + 3e^x - 10 &= 0\end{aligned}$$

Let $X = e^x$,

$$\begin{aligned}\therefore X^2 + 3X - 10 &= 0 \\(X + 5)(X - 2) &= 0 \\\therefore X = 2 \text{ or } -5\end{aligned}$$

If $X = 2$

$$\begin{aligned}e^x &= 2 \\\ln e^x &= \ln 2 \\x &= \ln 2\end{aligned}$$

If $X = -5$

$$e^x = -5 \quad \text{(no solution)}$$

$$\therefore x = \ln 2$$

28. L&E, 2ADV E1 2007 HSC 6a

Solution 1

$2e^{2x}-e^x=0$

Let $X=e^x$

$2X^2-X=0$

$X(2X-1)=0$

$X=0$ or $\frac{1}{2}$

When $e^x=0\Rightarrow$ no solution

When $e^x=\frac{1}{2}$

$\ln e^x=\ln \frac{1}{2}$

$\therefore x=\ln \frac{1}{2}$

Solution 2

$2e^{2x}-e^x=0$

$2e^{2x}=e^x$

$\ln 2e^{2x}=\ln e^x$

$\ln 2+\ln e^{2x}=x$

$\ln 2+2x=x$

$x=-\ln 2$

$=\ln \frac{1}{2}$

29. L&E, 2ADV E1 2008 HSC 7a

$\log_e x-\frac{3}{\log_e x}=2$

$(\log_e x)^2-3=2\log_e x$

$(\log_e x)^2-2\log_e x-3=0$

Let $X=\log_e x$

$\therefore X^2-2X-3=0$

$(X-3)(X+1)=0$

$X=3\qquad X=-1$

$\log_e x=3\qquad \log_e x=-1$

$x=e^3\qquad x=e^{-1}$

$\therefore x=e^3\text{ or }e^{-1}$

IMPORTANT: Students should recognise this equation as a quadratic, and the best responses substituted $\log_e x$ with a variable such as X .

MARKER'S COMMENT: Many responses incorrectly stated that there is no solution to $\log_e x=-1$ or could not find x given $\log_e x=3$.

30. L&E, 2ADV E1 2019 HSC 15a

$e^{2\ln x}=x+6$

$\ln e^{2\ln x}=\ln(x+6)$

$2\ln x=\ln(x+6)$

$\ln x^2=\ln(x+6)$

$x^2=x+6$

$x^2-x-6=0$

$(x-3)(x+2)=0$

$\therefore x=3\text{ (}x>0\text{)}$

♦ Mean mark 47%.

31. L&E, 2ADV E1 2020 MET1 4

$$\begin{aligned}2\log_2(x+5) - \log_2(x+9) &= 1 \\ \log_2(x+5)^2 - \log_2(x+9) &= 1 \\ \log_2\left(\frac{(x+5)^2}{x+9}\right) &= 1 \\ \frac{(x+5)^2}{x+9} &= 2 \\ x^2 + 10x + 25 &= 2x + 18 \\ x^2 + 8x + 7 &= 0 \\ (x+7)(x+1) &= 0 \\ \therefore x &= -1 \quad (x \neq -7 \text{ as } x > -5)\end{aligned}$$

32. L&E, 2ADV E1 EQ-Bank 2

$$\begin{aligned}\text{Odd} \Rightarrow f(-x) &= -f(x) \\ f(x) &= \frac{1}{2} - \frac{1}{2^x + 1} \\ f(-x) &= \frac{1}{2} - \frac{1}{2^{-x} + 1} \times \frac{2^x}{2^x} \\ &= \frac{1}{2} - \frac{2^x}{1 + 2^x} \\ &= \frac{1}{2} - \frac{2^x + 1 - 1}{2^x + 1} \\ &= \frac{1}{2} - 1 + \frac{1}{2^x + 1} \\ &= -\frac{1}{2} + \frac{1}{2^x + 1} \\ &= -f(x) \\ \therefore f(x) &\text{ is odd.}\end{aligned}$$

33. L&E, 2ADV E1 EQ-Bank 3

$$\begin{aligned}2^{2x} - 2^{x+2} &= 2^5 \\ 2^{2x} - 2^2 \times 2^x - 32 &= 0 \\ \text{Let } 2^x &= X \\ X^2 - 4X - 32 &= 0 \\ (X-8)(X+4) &= 0 \\ X = 8 \quad \text{or} \quad X &= -4 \\ 2^x = 8 \quad \quad 2^x &= -4 \text{ (no solution)} \\ x = 3 \\ \therefore x &= 3\end{aligned}$$

34. L&E, 2ADV E1 SM-Bank 15

$$\begin{aligned}2^{2x} &= 3(2^{x+1}) - 8 \\ 2^{2x} &= 3(2 \times 2^x) - 8 \\ 0 &= 2^{2x} - 6 \cdot 2^x + 8 \\ \text{Let } X &= 2^x \\ X^2 - 6X + 8 &= 0 \\ (X-4)(X-2) &= 0 \\ X = 4 \quad \text{or} \quad 2 \\ 2^x = 4 \Rightarrow x &= 2 \\ 2^x = 2 \Rightarrow x &= 1\end{aligned}$$

35. L&E, 2ADV E1 2016 HSC 14e

$$\begin{aligned}\log 2 + \log 4 + \log 8 + \dots + \log 512 \\ &= \log 2^1 + \log 2^2 + \log 2^3 + \dots + \log 2^9 \\ &= \log 2 + 2\log 2 + 3\log 2 + \dots + 9\log 2 \\ &= 45\log 2\end{aligned}$$

◆ Mean mark 40%.
TIP: Note that $\log 2 = \log_{10} 2$

36. L&E, 2ADV E1 2019 MET1-N 4

$$\log_3(t) - \log_3(t^2 - 4) = -1$$

$$\log_3\left(\frac{t}{t^2 - 4}\right) = -1$$

$$\frac{t}{t^2 - 4} = \frac{1}{3}$$

$$t^2 - 4 = 3t$$

$$t^2 - 3t - 4 = 0$$

$$(t - 4)(t + 1) = 0$$

$$\therefore t = 4 \quad (t > 0, t \neq -1)$$